

CENG381 Stochastic Processes

Martingales — Practice 1 (Answer Key)

Q1 Solution

a) Proving $\{S_n\}$ is a martingale:

$$\begin{aligned} & E[S_n \mid S_{n-1}, \dots, S_1] \\ &= E[S_{n-1} + X_n \mid S_{n-1}, \dots, S_1] \\ &= E[S_{n-1} \mid S_{n-1}, \dots, S_1] + E[X_n \mid S_{n-1}, \dots, S_1] \\ &= S_{n-1} + E[X_n] \quad (X_n \text{ independent of past}) \\ &= S_{n-1} + 0 = S_{n-1} \quad \checkmark \end{aligned}$$

The essential property is $E[X_i] = 0$. If the steps had non-zero mean the sum would drift and fail condition (ii).

b) Symmetric walk:

$$\begin{aligned} E[S_n \mid S_{n-1}] &= S_{n-1} + E[X_n] = S_{n-1} + (1/2 - 1/2) = S_{n-1} \quad \checkmark \\ E[S_n] &= E[S_0] = 0 \quad \text{for all } n \quad (\text{martingale has constant mean}) \end{aligned}$$

c) Biased walk:

$$\begin{aligned} E[X_i] &= p \cdot 1 + q \cdot (-1) = p - q = 2p - 1 \\ E[S_n \mid S_{n-1}, \dots, S_1] &= S_{n-1} + (2p-1) \\ p > 1/2: \quad E[S_n \mid \text{past}] &= S_{n-1} + (2p-1) > S_{n-1} \\ &\Rightarrow \text{sub-martingale (expected value tends to increase)} \\ p < 1/2: \quad E[S_n \mid \text{past}] &= S_{n-1} + (2p-1) < S_{n-1} \\ &\Rightarrow \text{super-martingale (expected value tends to decrease)} \end{aligned}$$

Q2 Solution

a) Proving $\{Z_n = S_n^2 - n\}$ is a martingale:

$$\begin{aligned} & E[Z_n \mid S_{n-1}] \\ &= E[S_n^2 - n \mid S_{n-1}] \\ &= E[(S_{n-1} + X_n)^2 \mid S_{n-1}] - n \\ &= E[S_{n-1}^2 + 2S_{n-1}X_n + X_n^2 \mid S_{n-1}] - n \end{aligned}$$

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$$\begin{aligned} &= S_{\{n-1\}}^2 + 2*S_{\{n-1\}}*E[X_n] + E[X_n^2] - n \\ &= S_{\{n-1\}}^2 + 2*S_{\{n-1\}}*0 + 1 - n \quad (E[X_n^2]=1 \text{ since } X_n=\pm 1) \\ &= S_{\{n-1\}}^2 - (n-1) \\ &= Z_{\{n-1\}} \quad \checkmark \end{aligned}$$

b) OST applied to $\{Z_n\}$, exit time T :

$$\begin{aligned} E[Z_T] &= Z_0 = S_0^2 - 0 = 0 \\ Z_T &= S_T^2 - T, \quad \text{and } |S_T| = N \Rightarrow S_T^2 = N^2 \\ E[N^2 - T] &= 0 \\ N^2 - E[T] &= 0 \\ E[T] &= N^2 \end{aligned}$$

c) Symmetry argument and $E[T]$ for $N=4$:

$$\begin{aligned} \text{OST on } \{S_n\}: E[S_T] &= S_0 = 0 \\ \text{Let } \alpha &= P(S_T = +N). \\ E[S_T] &= \alpha*N + (1-\alpha)*(-N) = 0 \\ \Rightarrow \alpha*N &= (1-\alpha)*N \Rightarrow \alpha = 1/2 \quad \checkmark \\ \text{For } N = 4: E[T] &= 4^2 = 16 \text{ rounds} \end{aligned}$$

Q3 Solution

a) Ruin probability via martingale OST:

$$\begin{aligned} E[S_T] &= S_0 = s \\ S_T &= N \text{ w.p. } p_s, \quad S_T = 0 \text{ w.p. } 1-p_s \\ p_s * N + (1-p_s) * 0 &= s \\ p_s &= s/N \end{aligned}$$

b) Expected game length via quadratic martingale OST:

$$E[Z_T] = Z_0 \Rightarrow E[S_T^2 - T] = s^2$$

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$$E[S_T^2] = E[T] + s^2$$

$$E[S_T^2] = p_s * N^2 + (1-p_s) * 0^2 = (s/N) * N^2 = s*N$$

$$s*N = E[T] + s^2$$

$$E[T] = s*N - s^2 = s*(N-s)$$

c) Gambler starting at \$3, goal \$7:

$$s = 3, \quad N = 7$$

$$P(\text{reach } 7) = p_3 = 3/7 \approx 0.429$$

$$E[T] = 3*(7-3) = 3*4 = 12 \text{ rounds}$$

The gambler has a 43% chance of reaching the goal and expects the game to end in 12 rounds on average.